

Fundamentals of Data science

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1. Sorting

Sorting is the process of arranging data in a specific order, such as ascending or descending order, based on one or more values or attributes.

Example: Arranging students according to their marks from highest to lowest.

2. Grouping

Grouping is the process of organizing data into categories or groups based on a common attribute. It helps in analyzing data within specific categories.

Example: Grouping students according to their departments such as CSE, ECE, and IT.

3. Ranking

Ranking is the process of assigning a position or order to data items based on their values or performance.

Example: Assigning Rank 1 to the student with the highest marks.

4. Plotting

Plotting is the graphical representation of data using charts, graphs, or visualizations. It helps in understanding patterns, trends, and relationships in data.

Example: Representing student marks using a bar chart or line graph.

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Name': ['Arun', 'Bala', 'Chitra', 'Divya', 'Ezhil', 'Fathima'],
    'Department': ['CSE', 'ECE', 'CSE', 'ECE', 'CSE', 'ECE'],
    'Marks': [85, 78, 92, 88, 75, 95]
}
```

```
df = pd.DataFrame(data)
```

```
print("Original Data:")
print(df)
```

1. SORTING

```
sorted_df = df.sort_values(by='Marks', ascending=False)
print(sorted_df)
```

```
1.Hand on work Sorting.py - C:/Users/admin/Downloads/review 2 geotag/1.Hand on work Sorting.py (3.14.3)
File Edit Format Run Options Window Help

import pandas as pd

data = {
    "Name": ["Arun", "Bala", "Chitra", "Divya", "Ezhil", "Fathima"],
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE", "ECE"],
    "Marks": [85, 78, 92, 88, 75, 95]
}

df = pd.DataFrame(data)

sorted_df = df.sort_values(by="Marks", ascending=False)

print(sorted_df)
```

```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/1.Hand on work Sorting.py =
      Name Department Marks
5  Fathima         ECE    95
2   Chitra         CSE    92
3   Divya         ECE    88
0    Arun         CSE    85
1    Bala         ECE    78
4    Ezhil         CSE    75

>>>
```

2. GROUPING

```
grouped_df = df.groupby('Department')['Marks'].mean()
print(grouped_df)
```

```
2.Hand on work grouping.py - C:/Users/admin/Downloads/review 2 geotag/2.Hand on work grouping.py (3.14.3)
File Edit Format Run Options Window Help

import pandas as pd

data = {
    "Name": ["Arun", "Bala", "Chitra", "Divya", "Ezhil", "Fathima"],
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE", "ECE"],
    "Marks": [85, 78, 92, 88, 75, 95]
}

df = pd.DataFrame(data)

grouped_df = df.groupby("Department")["Marks"].mean()

print(grouped_df)
```

```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help

Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
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>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/1.Hand on work Sorting.py =
      Name Department Marks
5  Fathima         ECE    95
2   Chitra         CSE    92
3   Divya         ECE    88
0    Arun         CSE    85
1    Bala         ECE    78
4    Ezhil         CSE    75

>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/2.Hand on work grouping.py =
Department
CSE      84.0
ECE      87.0
Name: Marks, dtype: float64

>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/2.Hand on work grouping.py =
Department
CSE      84.0
ECE      87.0
Name: Marks, dtype: float64

>>>
```

3. RANKING

```
df['Rank'] = df['Marks'].rank(ascending=False, method='dense')
print(df.sort_values(by='Rank'))
```

```
3.Hand on work Ranking.py - C:/Users/admin/Downloads/review 2 geotag/3.Hand on work Ranking.py (3.14.3)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    "Name": ["Arun", "Bala", "Chitra", "Divya", "Ezhil", "Fathima"],
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE", "ECE"],
    "Marks": [85, 78, 92, 88, 75, 95]
}

df = pd.DataFrame(data)

df["Rank"] = df["Marks"].rank(ascending=False, method="dense")

print(df.sort_values(by="Rank"))
```

```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/1.Hand on work Sorting.py ==
   Name Department Marks
5  Fathima      ECE    95
2  Chitra      CSE    92
3  Divya      ECE    88
0  Arun      CSE    85
1  Bala      ECE    78
4  Ezhil      CSE    75

>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/2.Hand on work grouping.py ==
Department
CSE    84.0
ECE    87.0
Name: Marks, dtype: float64

>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/2.Hand on work grouping.py ==
Department
CSE    84.0
ECE    87.0
Name: Marks, dtype: float64

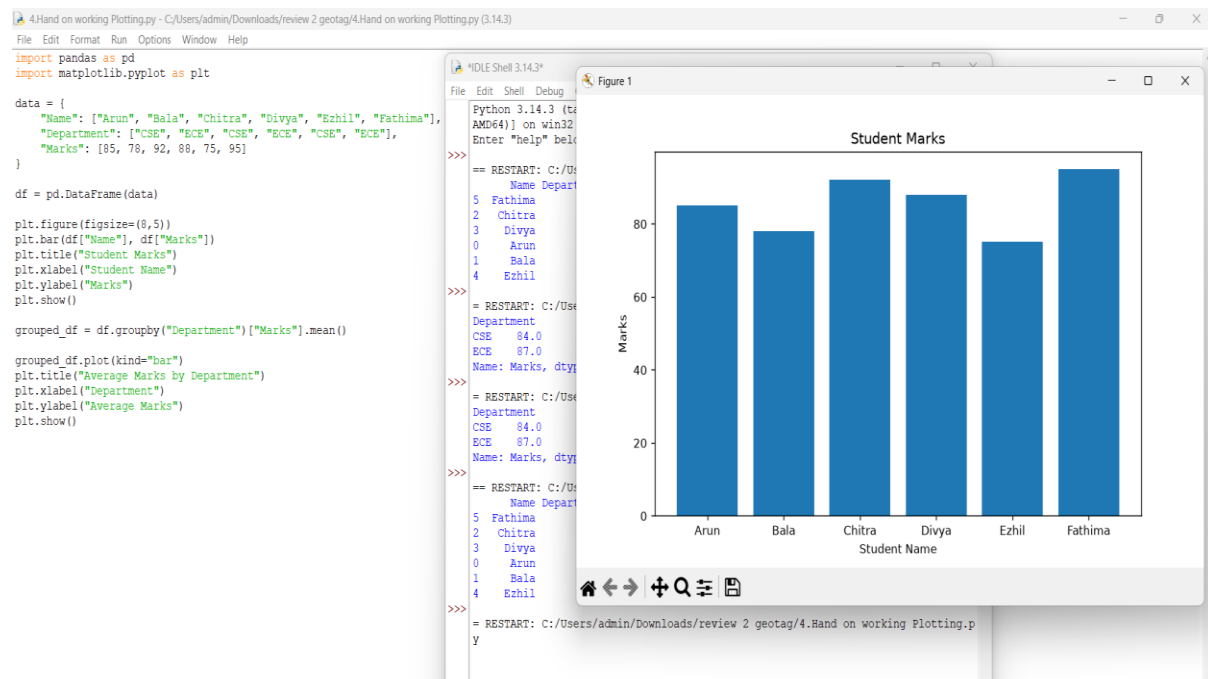
>>>
== RESTART: C:/Users/admin/Downloads/review 2 geotag/3.Hand on work Ranking.py ==
   Name Department Marks Rank
5  Fathima      ECE    95  1.0
2  Chitra      CSE    92  2.0
3  Divya      ECE    88  3.0
0  Arun      CSE    85  4.0
1  Bala      ECE    78  5.0
4  Ezhil      CSE    75  6.0

>>>
```

4. PLOTTING

```
# Bar chart of students' marks
plt.figure(figsize=(8, 5))
plt.bar(df['Name'], df['Marks'])
```

```
plt.title('Student Marks')
plt.xlabel('Student Name')
plt.ylabel('Marks')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Plot average marks by department

```
grouped_df.plot(kind='bar')
```

```
plt.title('Average Marks by Department')
```

```
plt.xlabel('Department')
```

```
plt.ylabel('Average Marks')
```

```
plt.xticks(rotation=0)
```

```
plt.tight_layout()
```

```
plt.show()
```

4.Hand on working Plotting.py - C:/Users/admin/Downloads/review 2 geotag/4.Hand on working Plotting.py (3.14.3)

File Edit Format Run Options Window Help

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
data = {
    "Name": ["Arun", "Bala", "Chitra", "Divya", "Ezhil", "Fathima"],
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE", "ECE"],
    "Marks": [85, 78, 92, 88, 75, 95]
}
```

```
df = pd.DataFrame(data)
```

```
plt.figure(figsize=(8,5))
plt.bar(df["Name"], df["Marks"])
plt.title("Student Marks")
plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.show()
```

```
grouped_df = df.groupby("Department")["Marks"].mean()
```

```
grouped_df.plot(kind="bar")
plt.title("Average Marks by Department")
plt.xlabel("Department")
plt.ylabel("Average Marks")
plt.show()
```

"IDLE Shell 3.14.3"

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Python 3.14.3 (tags/v3.14.3:2556e88, Apr 20, 2024) on win32
Enter "help" below or c

```
>>> == RESTART: C:/Users/ad
```

```
      Name Department
5  Fathima      ECE
2   Chitra      CSE
3   Divya      ECE
0    Arun      CSE
1    Bala      ECE
4   Ezhil      CSE
```

```
>>> == RESTART: C:/Users/ad
```

```
      Department
CSE      84.0
ECE      87.0
Name: Marks, dtype: flo
```

```
>>> == RESTART: C:/Users/ad
```

```
      Department
CSE      84.0
ECE      87.0
Name: Marks, dtype: flo
```

```
>>> == RESTART: C:/Users/ad
```

```
      Name Department
5  Fathima      ECE
2   Chitra      CSE
3   Divya      ECE
0    Arun      CSE
1    Bala      ECE
4   Ezhil      CSE
```

```
>>> == RESTART: C:/Users/ad
```

```
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y
```

